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DataBase System

Cake Master - GUI

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Project Proposal

1. INTRODUCTION

The Cake Management System is a Java-based desktop application with MySQL database integration designed to automate bakery operations. The system provides a graphical user interface (GUI) for both administrators and customers, enabling efficient management of cake inventory, customer orders, payments, deliveries, and feedback with persistent data storage. This project upgrades the previously developed console-based OOP version to a fully functional database-driven application.

2. PROBLEM STATEMENT

Bakeries face significant challenges in manually managing their daily operations, including:

- ✧ Tracking cake inventory and stock levels
- ✧ Processing customer orders efficiently
- ✧ Maintaining customer records and order history
- ✧ Managing payments and delivery coordination
- ✧ Monitoring delivery personnel

Manual processes and console-based systems often lead to errors, delays, and data loss. There is a clear need for an automated system with **database persistence** and a **graphical user interface**.

3. OBJECTIVE

3.1 Primary Objectives

Objective	Description
Inventory Management	Maintain complete cake catalog with flavors, prices, and sizes
Order Processing	Enable customers to place, track, and cancel orders

Objective	Description
Customer Management	Store and manage customer information with order history
Payment Processing	Support EasyPaisha, JazzCash, and Cash on Delivery
Delivery Management	Track delivery personnel and delivery status
Feedback System	Collect customer feedback and ratings (1-5 stars)
Data Persistence	Store all data securely in MySQL database

3.2 Technical Objectives

- ✧ Implement Object-Oriented Programming principles
- ✧ Develop Java application with Swing GUI
- ✧ Integrate MySQL database using JDBC
- ✧ Implement role-based access control (Admin and Customer)

4. SCOPE

4.1 Entities Managed

Entity	Key Attributes
Customer	customer_id, username, password, name, email, mobile_no, role
Cake	cake_flavor, price, size
Admin Order	order_id, customer_id, customer_name, order_status
Customer Order	order_id, customer_id, cake_type, size, quantity, price, rating, feedback
Customer Payment	order_id, mobile_no, payment_type, payment_status
Delivery Address	order_id, house_no, street_no, city, date, day, time
Delivery Personnel	personal_id, personal_name, order_id, email, mobile_no, status

4.2 Core Operations

Module	Operations
Customer	Sign up, login, browse menu, place orders, track orders, payment, feedback
Admin	Manage customers, catalog, orders, delivery personnel, generate reports

5. SYSTEM ARCHITECTURE

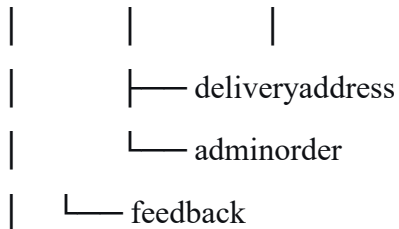
5.1 Technology Stack

Component	Technology
Language	Java (JDK 8+)
Database	MySQL
GUI Framework	Java Swing
JDBC Driver	MySQL Connector/J
Paradigm	Object-Oriented Programming

5.1 Database Schema

The system uses **7 interconnected tables**:

customer — customerorder — customerpayment



All tables are normalized

Primary and foreign key relationships maintained

Indexes for performance optimization

6. Features

6.1 Customer Features

Feature	Description
Sign Up	Register with name, email, mobile, username, password
Log In	Secure login with username and password
Browse Cake Menu	View all available cakes with prices and sizes
Place Order	Select cake, size, and quantity
Track Order	View order status and details
Track Delivery	Enter address and track delivery status
Payment Options	EasyPaisa, JazzCash, or Cash on Delivery
Feedback & Ratings	Submit reviews and rate orders (1-5 stars)
Cancel Order	Cancel orders before delivery

6.2 Admin Features

Feature	Description
Admin Login	Secure login with credentials
Customer Access	Add, view, update, delete customer profiles
Catalog Management	Add, update, remove, display cakes
Order Management	Add, track, update status, delete orders
Financial Reporting	Generate sales reports
Delivery Control	Add personnel, update status, view personnel list

7. Comparison

Feature	OOP Version (v1.0)	Database Version (v2.0)
Storage	Collections (ArrayList)	MySQL Database
Interface	Console-based	Swing GUI
Data Persistence	✗ Lost on exit	✓ Permanent
GUI	✗ No	✓ Yes
Reports	✗ No	✓ Yes
User Experience	Basic	Interactive

8. SAMPLE TEST CASES

Test Case	Input	Expected Output
Add New Customer	Name, Email, Mobile, Username, Password	"Customer added successfully"
Add New Cake	Flavor, Price, Size	"Cake added successfully"
Place Order	Customer ID, Cake, Size, Quantity	Order confirmed with total price
Track Order	Order ID	Display order details with status
Make Payment	Order ID, Mobile No, Payment Type	"Payment confirmed"
Submit Feedback	Order ID, Rating, Feedback	"Thank you for your feedback"
Cancel Order	Order ID	"Order cancelled successfully"
Update Customer	Customer ID, New Details	"Customer updated successfully"
Generate Report	-	Display sales, profit/loss

9. LIMITATIONS

Limitation	Description
Platform	Desktop-based application
Database	Requires MySQL installation
Notifications	No automated email/SMS
GUI	Basic Swing interface

10. FUTURE IMPROVEMENTS

Enhancement	Description
Web Application	Convert to web-based system
Mobile App	Develop Android/iOS applications
Email Notifications	Send order confirmation and delivery updates
Payment Gateway	Integrate with real payment gateways
SMS Alerts	Delivery status updates via SMS

11. CONCLUSION

The Cake Management System provides a comprehensive solution for bakeries to manage their operations effectively. By automating inventory tracking, order processing, payment handling, and delivery management with **MySQL database persistence** and a **graphical user interface**, the system reduces manual errors and improves efficiency. The system demonstrates the integration of OOP principles, database connectivity, and GUI development.